Data Warehousing and Modeling

Lecture 4
Dimensional modeling

Dr. Jehan Y. Abu-Elreesh

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Dimensional modeling

- Like enterprise relationship (ER) modeling, **dimensional modeling** is a logical design technique.
- **Dimensional modeling** is much better suited for **business intelligence (BI) applications** and **data warehousing (DW)**.
- It depicts business processes throughout an enterprise and **organizes** that data and its **structure** in a logical way.
- The purpose of dimensional modeling is to **enable BI reporting, query, and analysis**.
- The key concepts in dimensional modeling are **facts**, **dimensions**, and **attributes**. There are different types of facts, depending on whether they can be added together

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Dimensional modeling

Dimensions can have different hierarchies, and have attributes that define the who, what, where, and why of the dimensional model. **The grain, or level of granularity**, is another key concept with dimensional modeling, as it determines the level of detail.

- **Facts**, **dimensions**, and **attributes** can be organized in several ways, called **schemas**. The choice of schema depends on **variables** such as the type of reporting that the model needs to facilitate and the type of BI tool being used.

- Building a dimensional model includes additional puzzle pieces such as **calendar** and **time dimensions**, and more complicated pieces such as **degenerative dimensions** and **consolidated fact tables**.

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Dimensional modeling

Facts

– A **fact** is a measurement of a business activity, such as a business event or transaction, and is generally numeric. Examples of facts are sales, expenses, and inventory levels. Numeric measurements may include counts, dollar amounts, percentages, or ratios.

– A fact is a collection of related data consisting of measures.

– In a data warehouse, facts are implemented in the core tables in which all of the numeric data is stored.

– Facts can be aggregated or derived. For example, you can sum up the total revenue or calculate the profitability of a set of sales transactions

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Dimensional modeling

Facts

– A **fact** is a measurement of Facts provide the measurements of how well or how poorly the business is performing.

– A fact is also referred to as organizational performance measure.

– Fact contains **redundancy**.

– **Ninety-percent** of the data in a dimensional model is typically located in the fact tables.

– The key dimensional modeling design concerns when working with the data in fact tables are how to minimize and standardize it and make it consistent.

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Dimensional modeling

Facts

– Fact tables are composed of two types of columns: **keys and measures**.

– The first, the key column, consists of a group **of foreign keys (FK)** that point to the primary keys of dimensional tables that are associated with this fact table to enable business analysis.

– The relationships between fact tables and the dimensions are one-to many.

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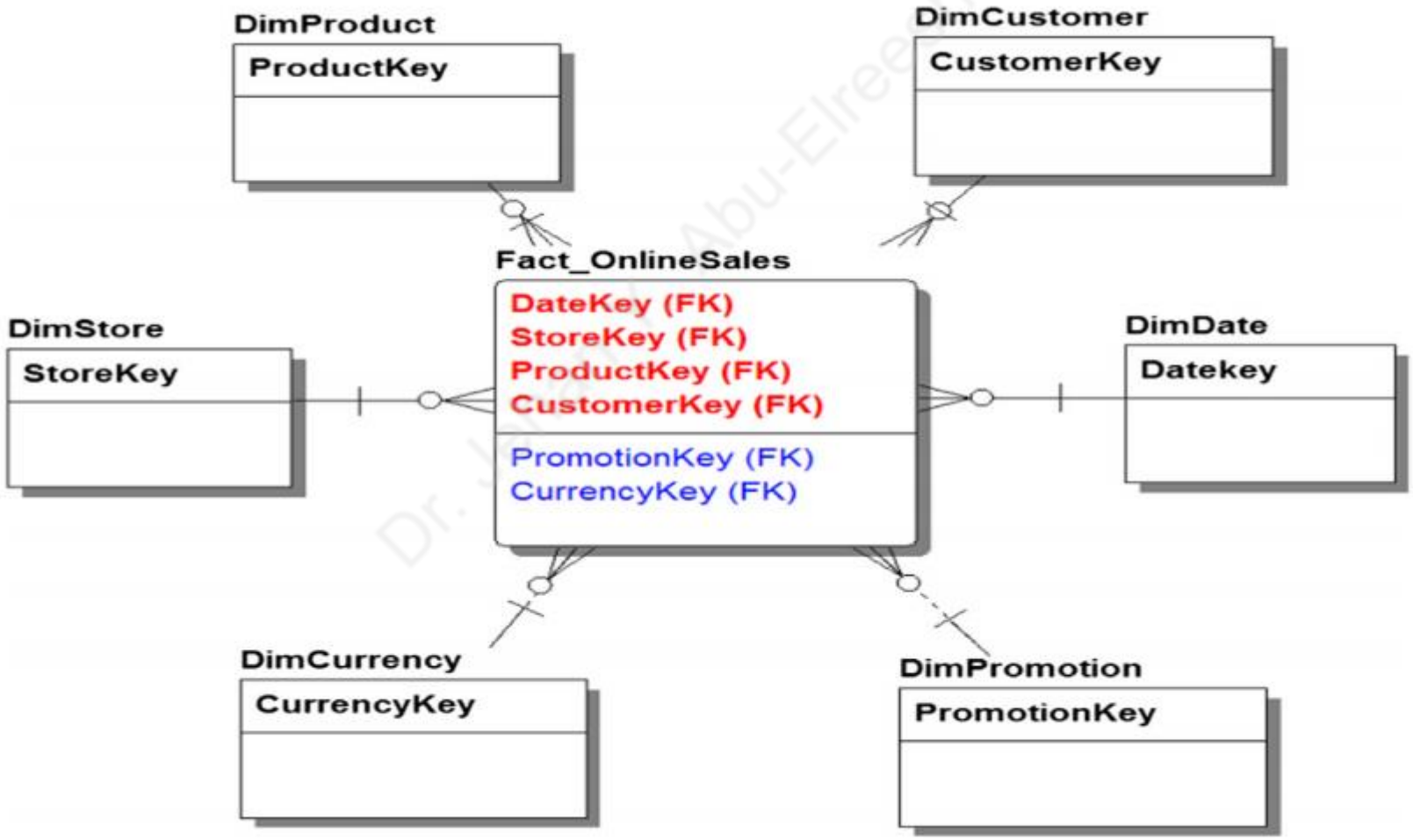
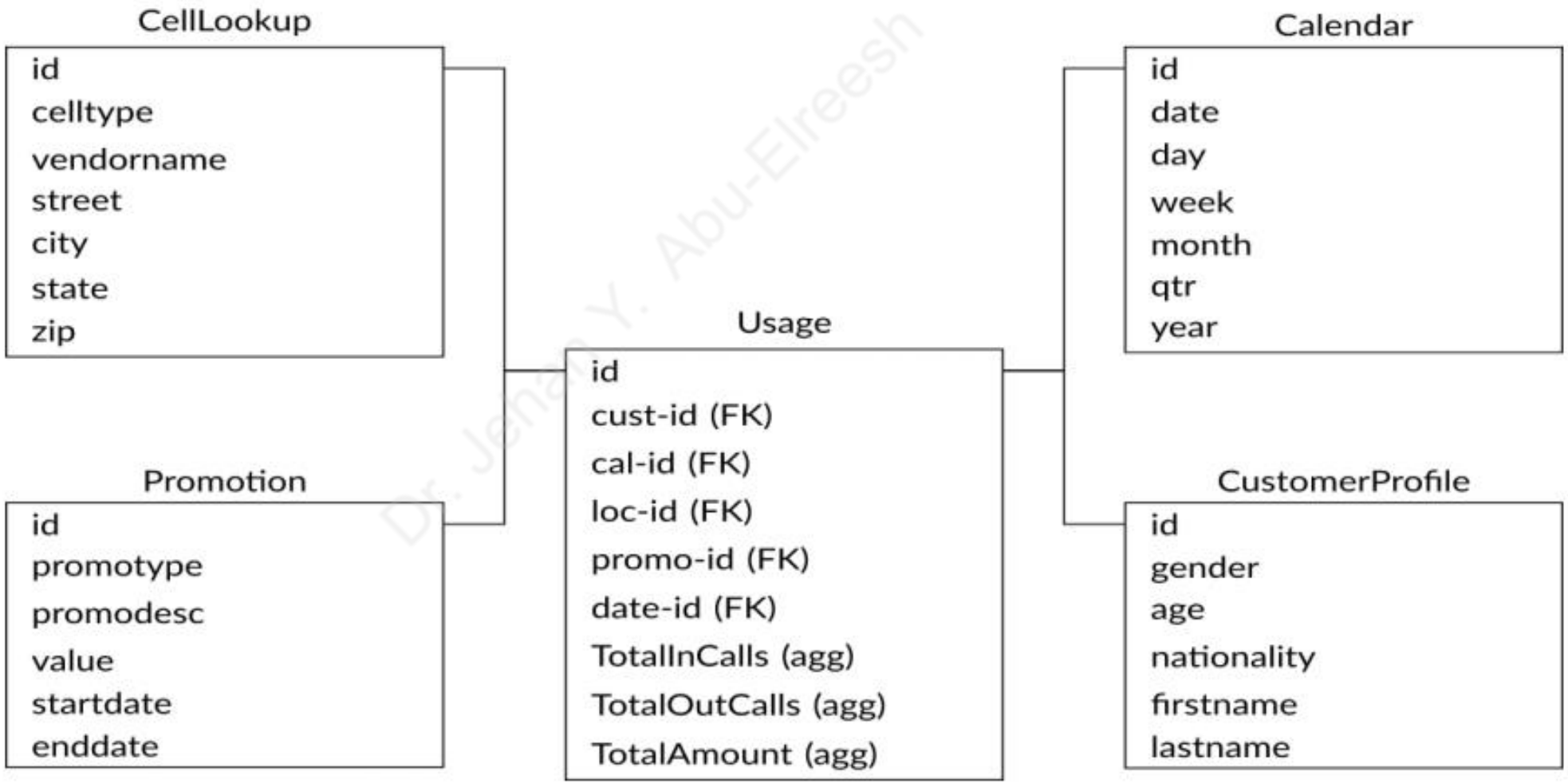
Dimensional modeling

Facts



Dimensional modeling

Facts



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Dimensional modeling

Facts – The second type of column in a fact table is the actual **measures** of the business activity such as the **sales revenue and order quantity.**

– Every measurement has a grain, which is the level of detail in the measurement of an event such as a unit of measure, currency used, or ending daily balance of an account.

– For example SalesQuantity, SalesAmount, ReturnAmount, ReturnQuantity, DiscountAmount, DiscountQuantity, and TotalCost that apply to a customer for a product purchased at a specific time.

– All of these measures are related to the business event (the sale) that the fact represents and they have a level of granularity related to that event.

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Dimensional modeling

Facts

Fact_OnlineSales	
PK	DateKey
PK	StoreKey
PK	ProductKey
PK	CustomerKey
	CurrencyKey
	PromotionKey
	SalesOrderNumber
	SalesOrderLineNumber
	SalesQuantity
	SalesAmount
	ReturnQuantity
	ReturnAmount
	DiscountQuantity
	DiscountAmount
	TotalCost
	UnitCost
	UnitPrice
	DW_ETLLoadID
	DW_LoadDate
	DW_UpdateDate

Dimensional modeling

Dimensions

– A dimension is an entity that establishes the business context for the measures (facts) used by an enterprise.

– Dimensions define the who, what, where, and why of the dimensional model, and group similar attributes into a category or subject area. Examples of dimensions are product, geography, customers, employees, and time.

– Whereas facts are **numeric**, dimensions are **descriptive** in nature (although some of those descriptions, such as a product list price, may be numeric).

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Dimensional modeling

Dimensions

DimProduct

ProductKey
ProductAltemateKey WeightUnitMeasureCode SizeUnitMeasureCode EnglishProductName StandardCost FinishedGoodsFlag Color SafetyStockLevel ReorderPoint ListPrice Size SizeRange

Schema Types

Schema types are :

1. Star schema

2. Snowflake schema

You choose which schema to use when building the dimensional model by considering these questions:

1. What kind of analysis are you trying to perform on that data and how complex is it?

2. What are the analytical requirements and restrictions?

3. How consistent is the data you want to query and analyze?

4. What BI tool do you plan to use? Although different tools may appear to show the same type of data, results, and graphs, they can be very different under the covers, and rely on a specific schema for the best results.

Note: most slides in this file produced from : Sharda, Ramesh, Dursun Delen, and Efraim Turban. Business intelligence, analytics, and data science: a managerial perspective. pearson, 2018.

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Schema Types

Star schema

- The star schema (sometimes referenced as star join schema) is the most commonly used and the simplest style of dimensional modeling.
- A star schema contains a central **fact table** surrounded by and connected to **several dimension** tables.
- The fact table contains a large number of rows that correspond to observed facts and external links (i.e., foreign keys).
- A fact table contains the descriptive attributes needed to perform decision analysis and query reporting, and foreign keys are used to link to dimension tables.

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Schema Types

Star schema

The decision analysis attributes consist of performance measures, operational metrics, aggregated measures (e.g., sales volumes, customer retention rates, profit margins, production costs, scrap rate), and all the other metrics needed to analyze the organization's performance.

- In other words, the **fact** table primarily addresses what the data warehouse supports for **decision analysis**.
- Surrounding the central fact tables (and linked via foreign keys) are **dimension tables.**

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Schema Types

Star schema

The dimension tables contain classification and aggregation information about the central fact rows.

- Dimension tables contain attributes that describe the data contained within the fact table; they address how data will be analyzed and summarized.
- Dimension tables have a **one-to-many** relationship with rows in the central **fact table**.
- In querying, the dimensions are used to **slice and dice** the numerical values in the fact table to address the requirements of an ad hoc information need.

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Schema Types

Star schema

Star Schema Main Characteristics

1. **Simplicity:** It is the simplest type of DWH schemas.

2. Query effectiveness: Because of simplicity, It needs less join to query the data (It is optimized to query large dataset).

3. Data Redundancy and Large Table Size: Due to denormalization, it has a data redundancy, and the table size is huge.

4. Most used and widely supported.

Note: This slide from: https://gability.com/en/courses/big-data-in-depth/02-dwh/03-architecture/10-data-modeling/04-schema-types/

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Schema Types

Star schema

Star schema additional Characteristics

1. Dimensions represented by one one-dimension table.

2. The dimension table are not joined to each other

3. The fact table would contain key and measure.

4. Data integrity is not enforced due to the de-normalized structure.

Note : This slide from : https://gability.com/en/courses/big-data-in-depth/02-dwh/03-architecture/10-data-modeling/04-schema-types/

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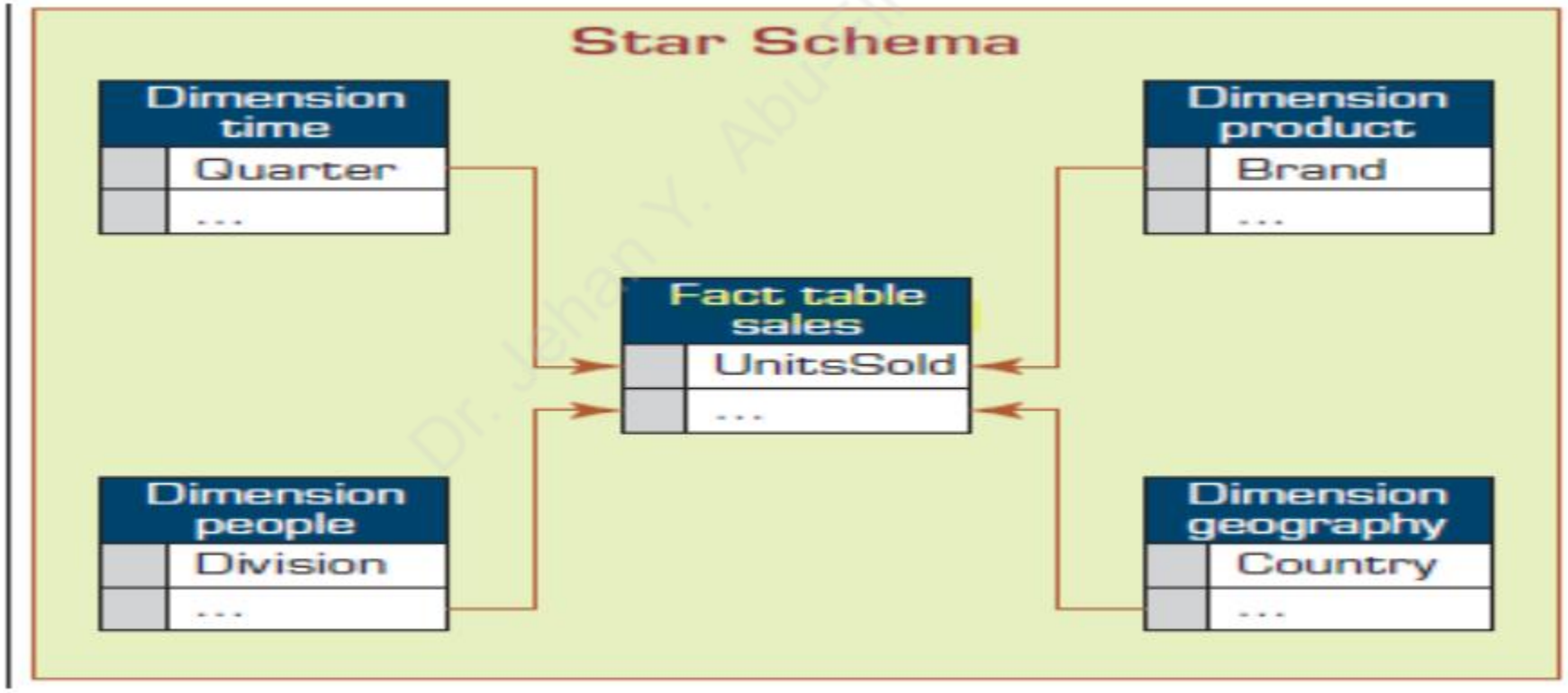
Schema Types

Star schema

The star schema is designed to provide fast query-response time, simplicity, and ease of maintenance for read-only database structure.



Schema Types

Snowflake schema

- Closely related to the star schema, the snowflake schema is represented by centralized fact tables (usually only one), which are connected to multiple dimensions.
- It takes the star schema, with the facts surrounded by denormalized dimensions, one step further by normalizing the hierarchies within a **particular dimension**.
- Each level in the dimensional hierarchy becomes its own dimensional table with parent keys created to link the hierarchical structure together.
- The fact table stores the foreign key to the lowest level of the dimensional hierarchy.

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Schema Types

Snowflake schema

- Dimensions are normalized into multiple related tables, whereas the star schema's dimensions are denormalized, with each dimension being represented by a single table.

Snowflake Schema Characteristics

1. Extension: Snowflake is an extension of the Star Schema.

2. Normalized: Dimension tables are normalized; this means every dimension may expand into additional tables.

3. Disk Space Efficiency: Due to its normalization methodology, it uses less desk space, which enhances the query as we scan less data size.

4. Complicated: Due to the normalization query needs to join more table in some cases to get the data which reduces the performance.

Dr. Jehad Y. Abu-Ereesh

Schema Types

Snowflake schema

Snowflake Schema Characteristics

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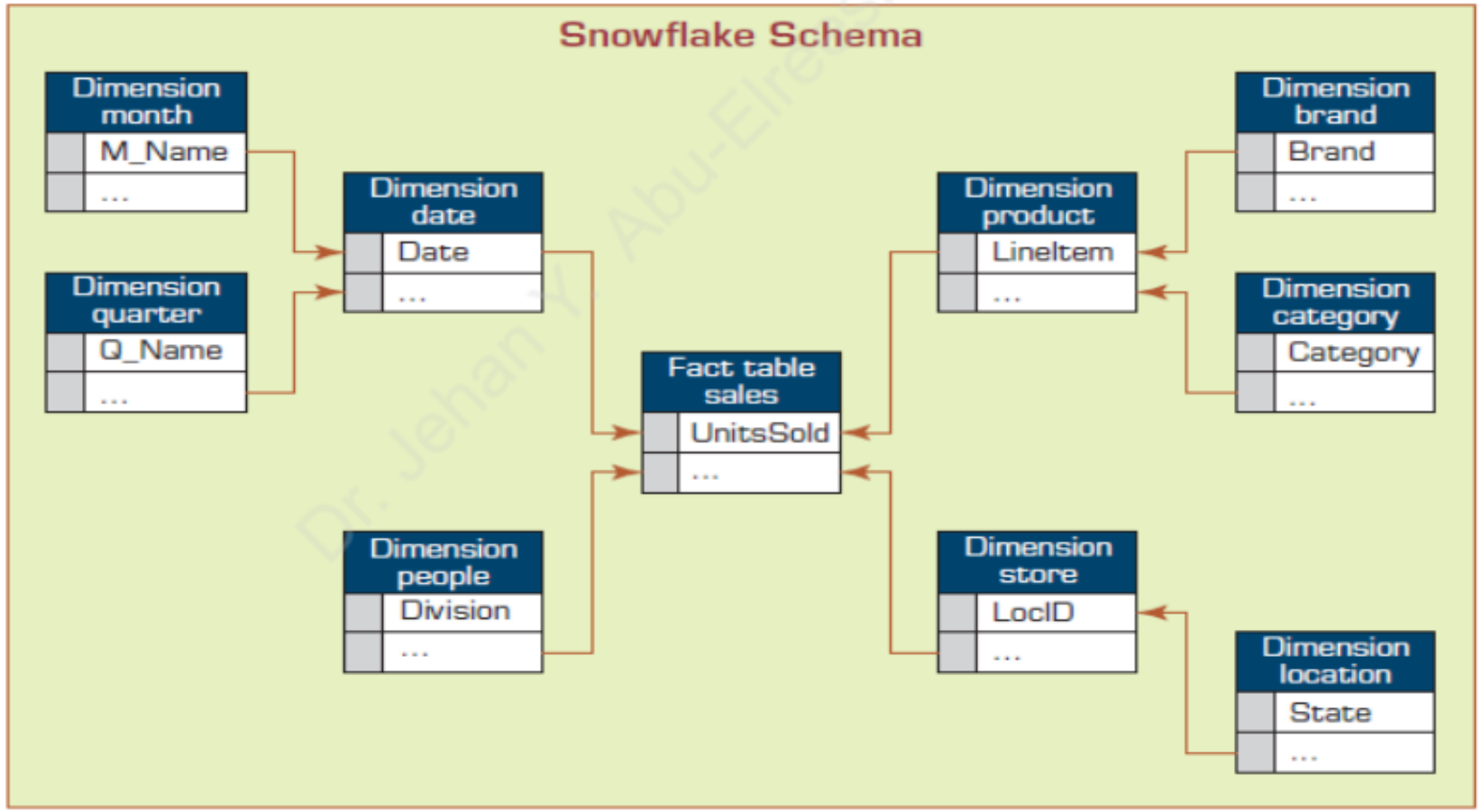
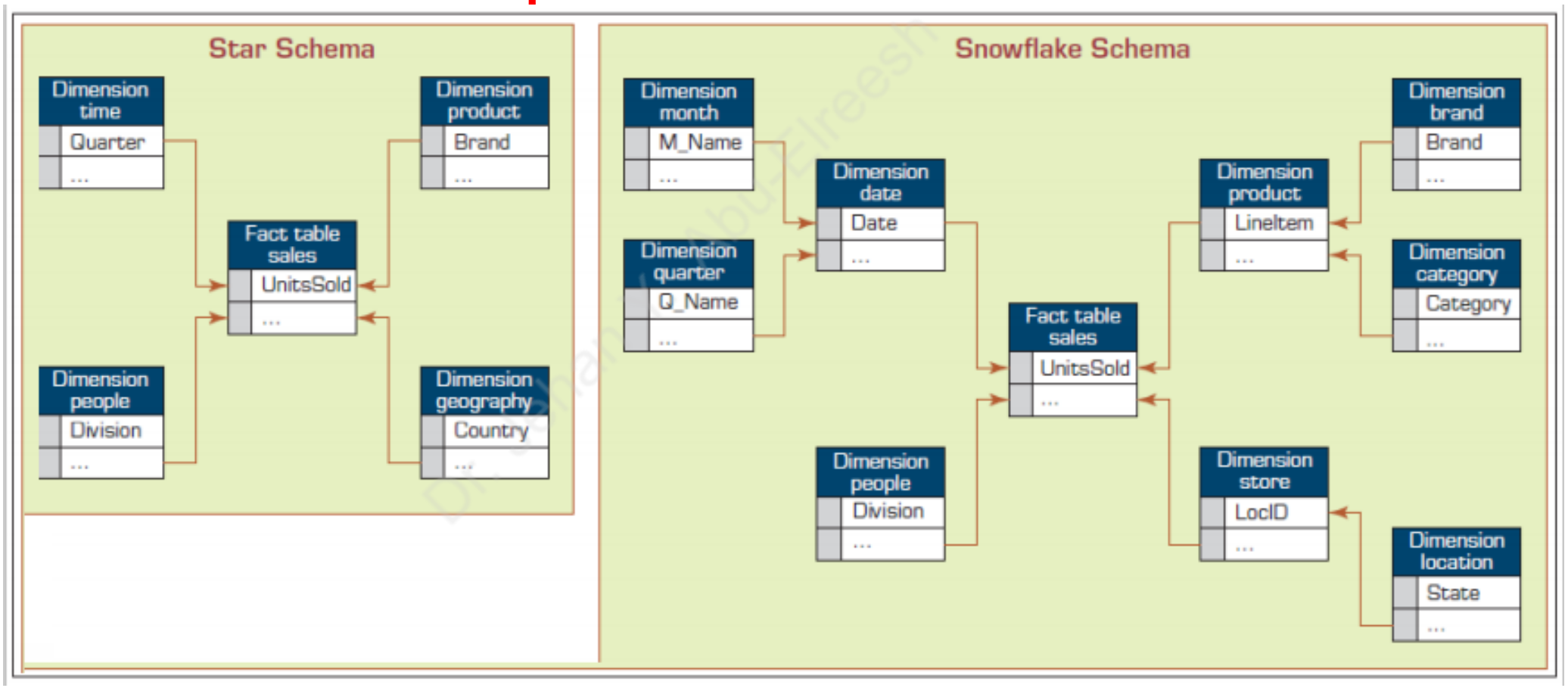
Schema Types

Snowflake schema



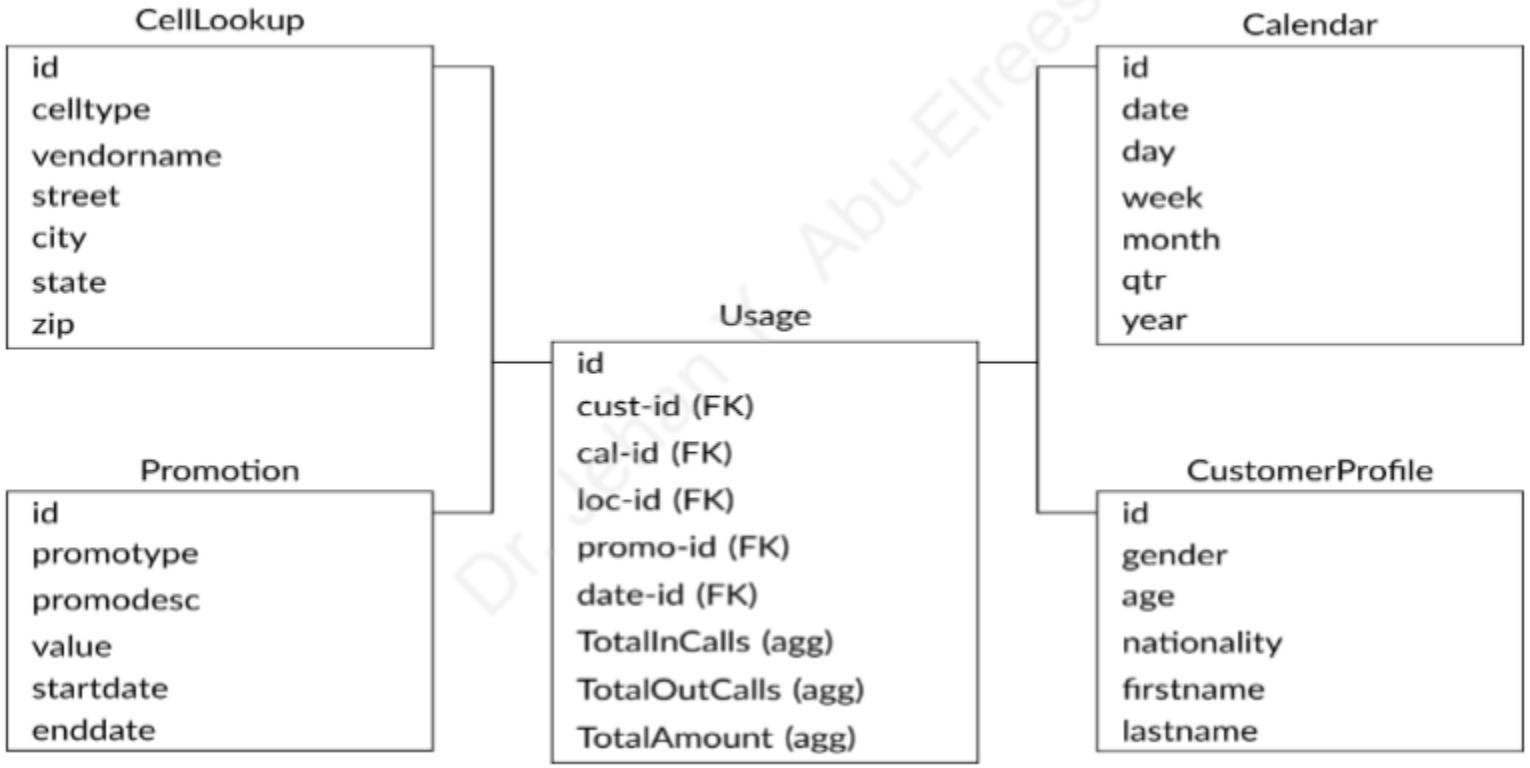
Schema Types

Star vs snowflake example



Schema Types

Star



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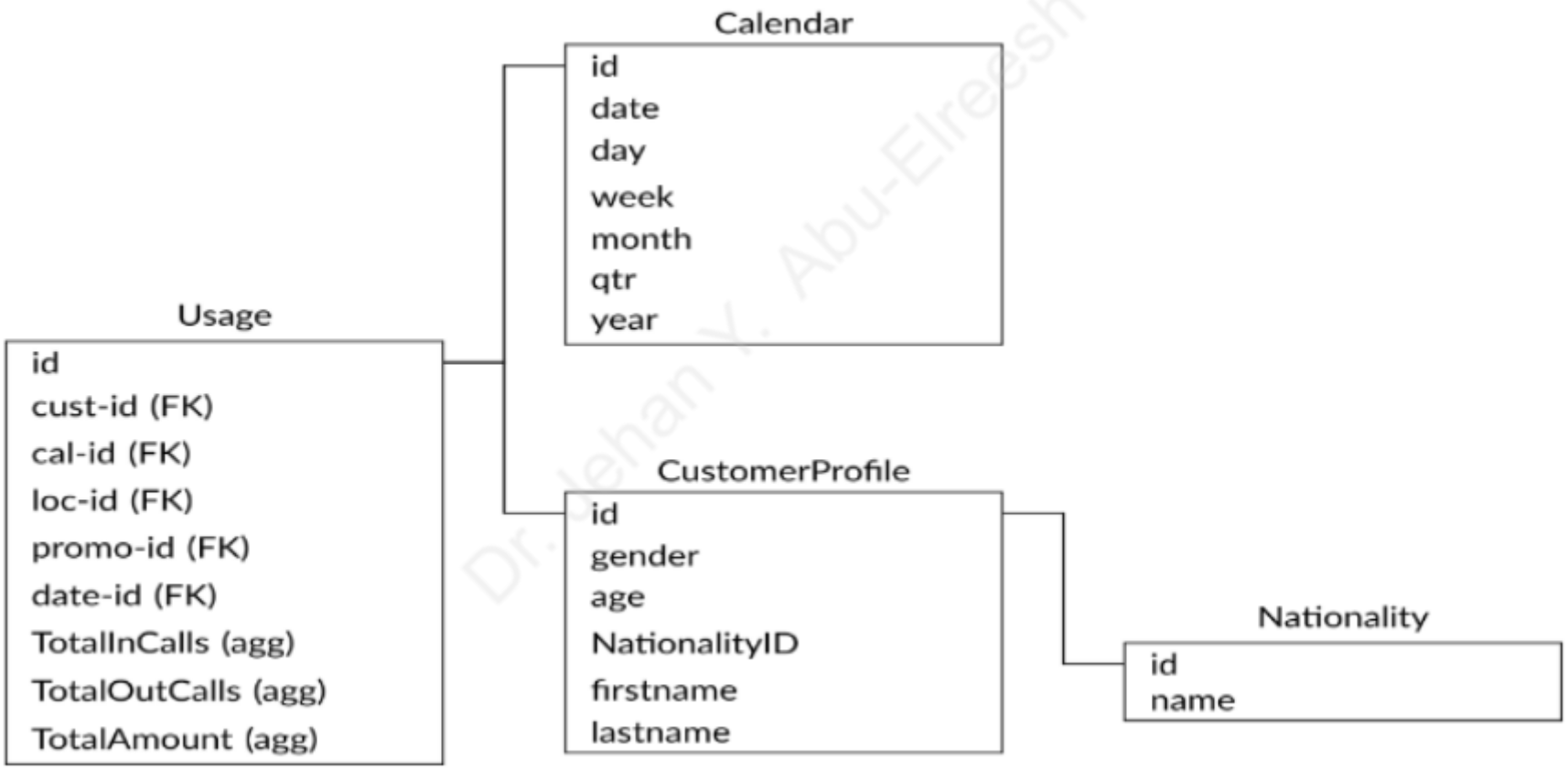
Schema Types

Snowflake



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Star	Snowflake
Dimension represented by one-table	Dimension tables are expanded into multi-tables
Fact table surrounded by dimension tables	Fact table surrounded by Hierarchy of dimension tables
Less join	Requires many joins
Simple Design	Very Complex Design
De-normalized Data structure	Normalized Data Structure
High level of Data redundancy	Very low-level data redundancy
Maintenance is difficult	Maintenance is easier
Good for data marts with simple relationships (1:1 or 1:many)	Good for core to simplify (many: many)

Dr. Jehan Y. Abu Elnasr

Schema Types

Multi-fact star models

- Examples of dimensional models, whether star, snowflake, or multidimensional, usually depict them with a single fact table surrounded by dimensions. Although this is a popular illustration and a terrific approach that simplifies explanations, the **reality is that data in an enterprise requires multiple facts.**
- Examples of facts are sales, expenses, inventory, and the many other business events associated with running an enterprise.
- For example , shows two facts: store sales and store inventory (Tbl_Fact_Store_Sales and Tbl_Fact_ Store_Inventory).

Dr. Jehad Y. Abu Elreesh

Schema Types

Multi-fact star models

- On the right side are three dimensions that both facts share: the item dimension, the date dimension, and the buyer dimension. The shared dimensions are referred to as **conformed dimensions.**
- On the left side of the example are the customer and store dimensions that are not shared across the two facts. The Tbl_Dim_Customer is related to Tbl_Fact_Store_Sales, whereas the Tbl_Dim_Store is related to Tbl_Fact_Store_Inventory

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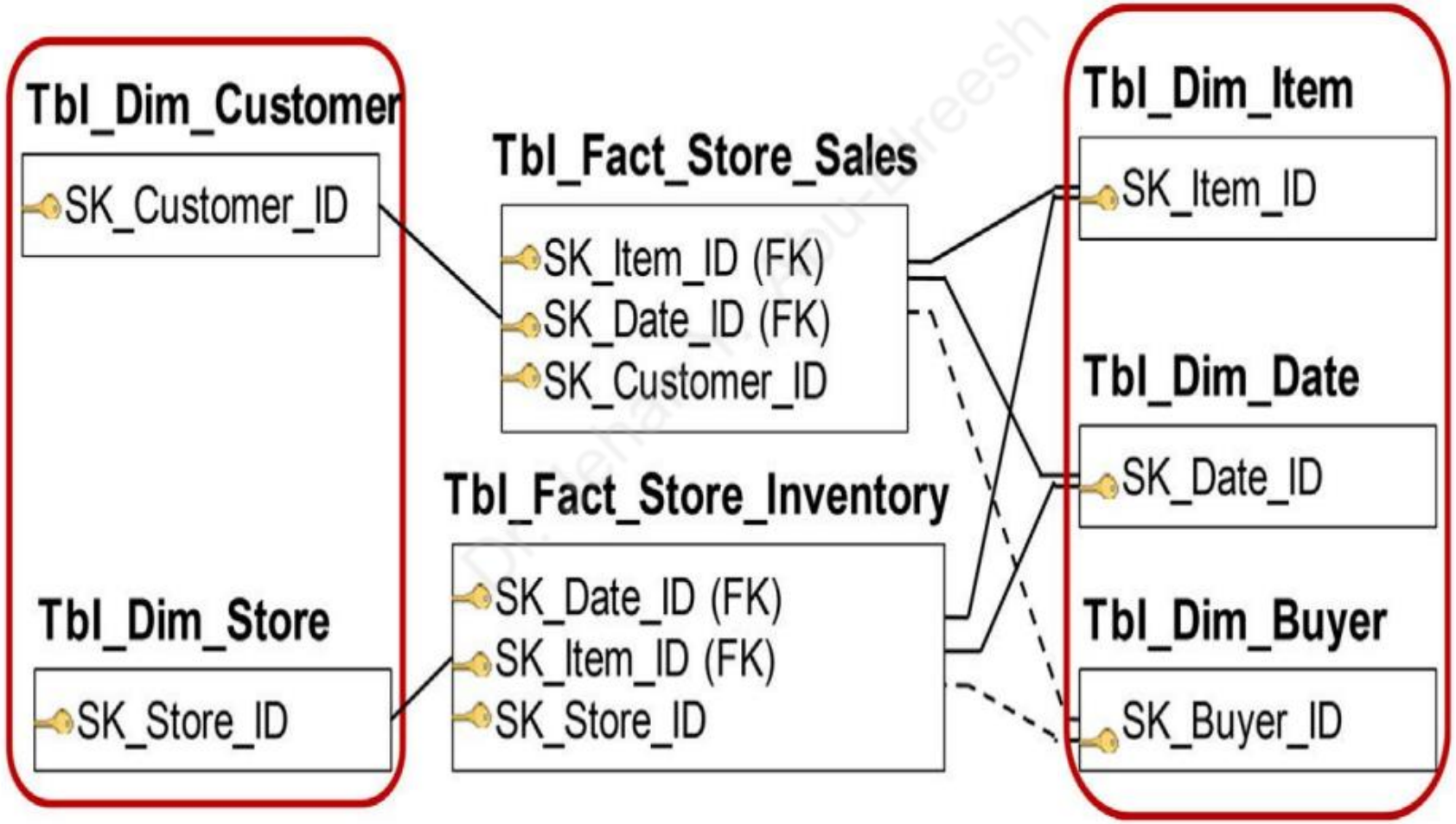
Schema Types

Multi-fact star models



ENTITY RELATIONSHIP vs DIMENSIONAL MODELING

ER modeling, sometimes referred to as normalized modeling, is the standard for transactional systems (also called operational or online transactional processing (**OLTP**) systems).

- Dimensional modeling is the best practice for BI and OLAP systems. The reason for this stems from how the two systems process and manipulate data.
- The data in operational systems(OLTP) is being constantly updated. The main job of these workhorses is transaction throughput. They have to maintain and update a large number of records, so they have to be able to handle high amounts of consistent throughput with large volumes.
- Compare that with the BI and reporting systems, which are generally not updating data. Rather, they are copying, extracting, transforming, and loading from operational systems. They are not doing realtime updates like operational systems.

Dr. Jehad I. Abu-Ellessi

ENTITY RELATIONSHIP vs DIMENSIONAL MODELING

The focus is not on transactional throughput once the data is loaded. The focus is on query performance, and gathering and aggregating large sets of data. That's a critical operation. Transactional systems update a small number of data records quite often, whereas BI systems gather and aggregate large sets of data records, but not so often—sometimes over large spans of time.

- The data from the BI system then get used for reports.
- The operational system and its accompanying normalized models have minimal **redundancy** to support the high transaction throughput.
- in BI reporting, there is a lot of **redundancy**. After all, the warehouse and data marts and whatever else you store on the reporting side are all copies of data that originally was in the operational systems. The BI system depends heavily on indexes and partitioning because it needs much more storage space for reporting.

Dr. Sohan Y. Abu-Firas

ENTITY RELATIONSHIP vs DIMENSIONAL MODELING

Operational Systems	BI and Analytics
Normalized models are standard for OLTP	Dimensional models are standard for BI and OLAP
Highly volatile updated	Generally not updated non-voltile
Transaction throughput (updating and maintaining numerous records) is critical	Query performance (gathering and aggregating large sets of records) is critical
Characteristics supporting use of normalized models:	**Characteristics supporting use of dimensional models:**
Minimal redundancy (normalization)	Increased redundancy (denormalization)
Limited index use	Increased index use
Efficient use of storage space	Increased storage space
Eliminate inconsistent data	Consolidate inconsistent data
Few maintenance concerns	Increased maintenance issues

3-Tires Data Warehouse Architecture

- **Middle Tier**

In the middule tier, we have the OLAP server that can be implemneted in either of the following ways:

1- By Relational OLAP-ROLAP, which is an extended relationl database management system. The ROLAP maps the operations on multidimensional data to standared relational operations.

2- By multidimensional OLAP-MOLAP model, which directly implements the multudemnetinal data and operations.

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Qustions ??
Thank You

The next Lecture will be :

Dimensional modeling (Facts Types)